

8.10.2025

Language

Closure Properties

Models of Computation

TM

- a. λ -calculus
- b. Term Rewriting
- c. Rec. Fn. Theory

Church
Turing
thesis

→ NP-completeness -

$$\underline{P \stackrel{?}{=} NP}$$

Quiz 2 on 15/10/25

→ Assign 4.

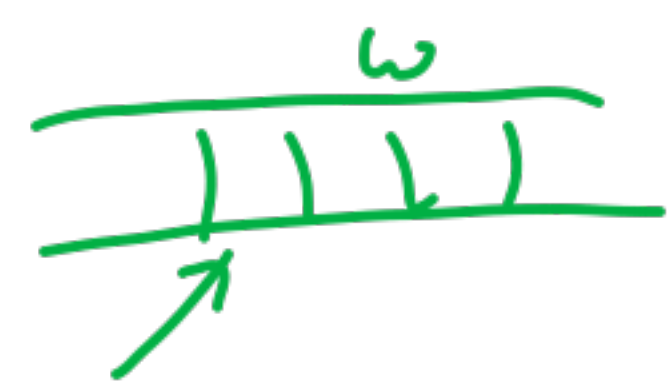
$\text{r.e. } L \text{ is rec En.} \iff \text{TM } M \text{ accepts } L$

$L = \{ ww \mid w \in (0+1)^* \}$

$\text{rec. Recursive} \iff \text{halts in reject state if } w \notin L$

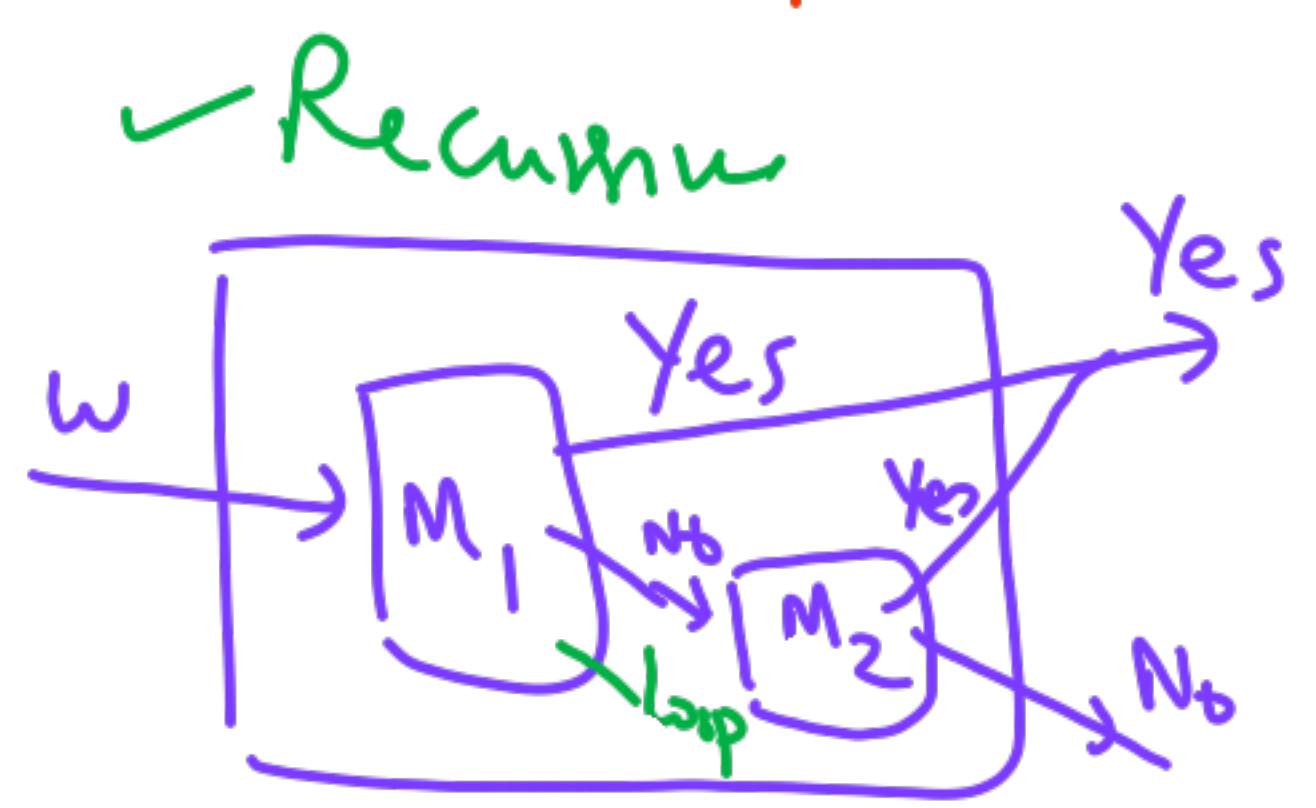
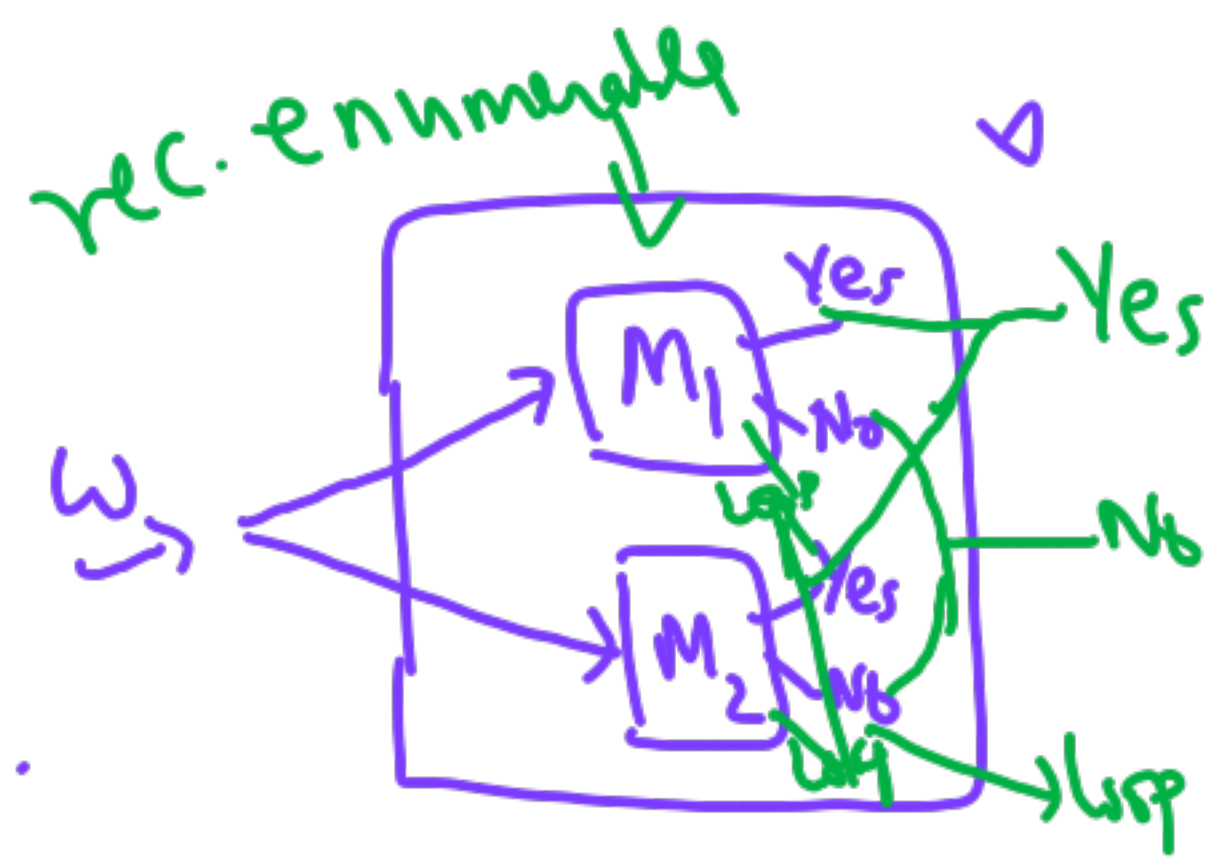
$\text{TM } M$ accepts L

$\text{halts in reject state if } w \notin L$



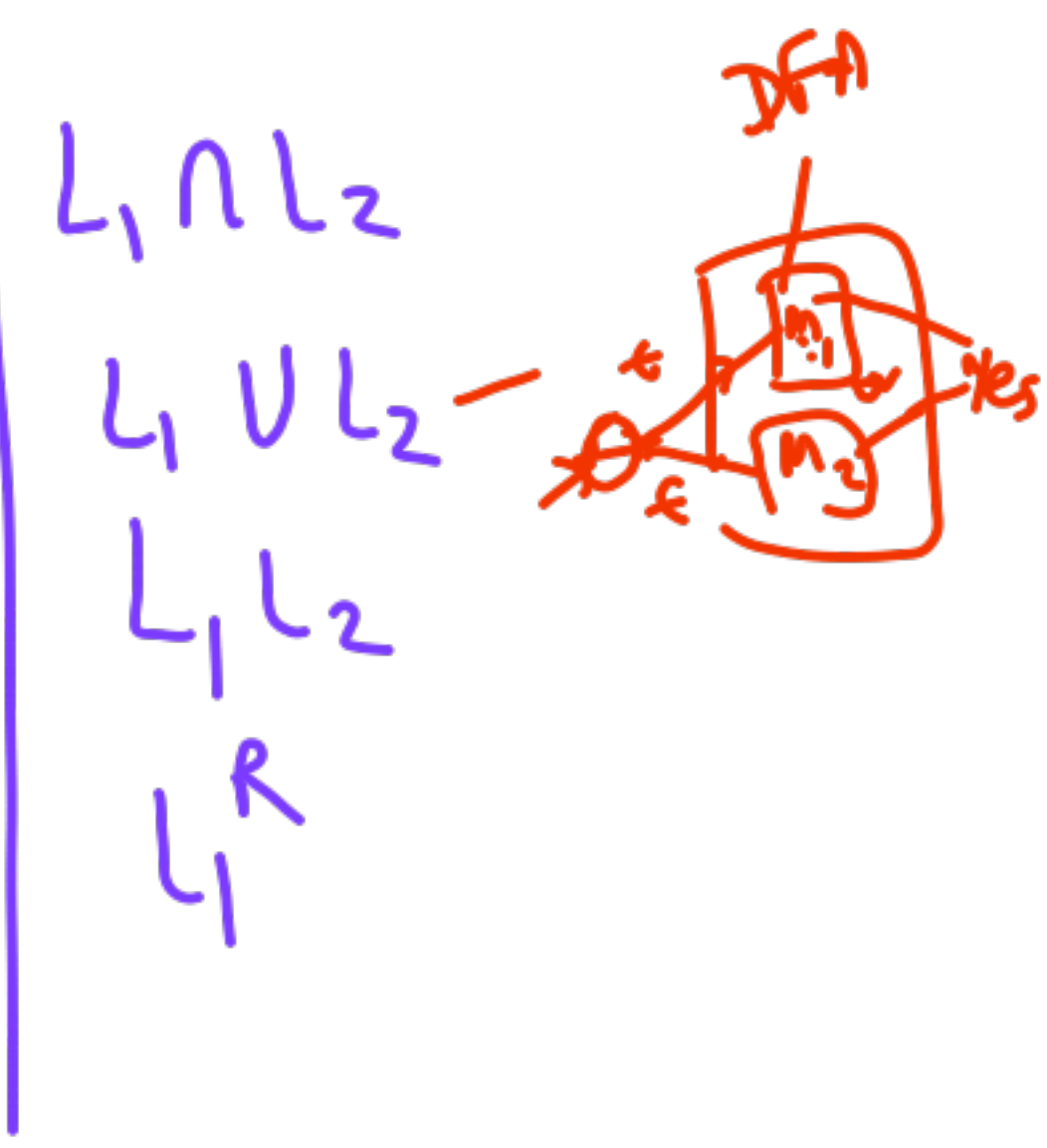
DFA/NFA
 $\uparrow$
 $L_1, L_2 \text{ are regular}$

Union L_1, L_2 are r.e. / rec. what about $L_1 \cup L_2$?



$\uparrow$
 allowed 3 behaviours

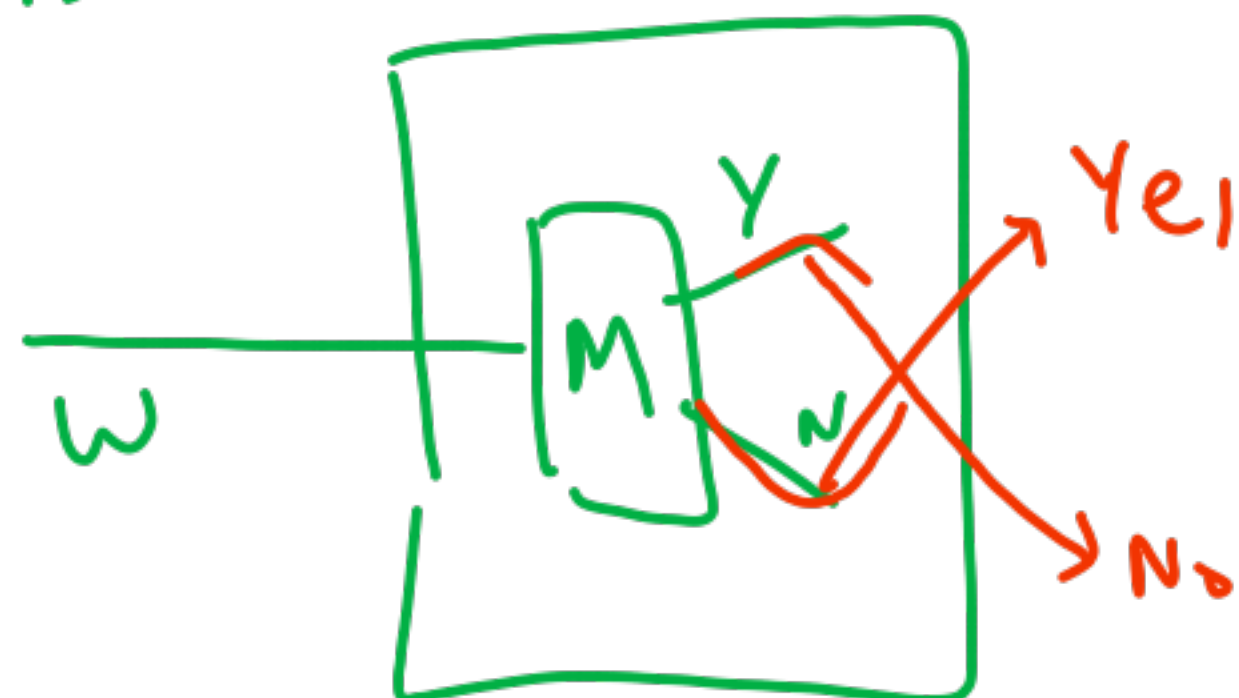
$\uparrow$
 only 2 behaviours



Complement

r.e. h/w

Rec
 L is rec - L^c ?



flip
answer

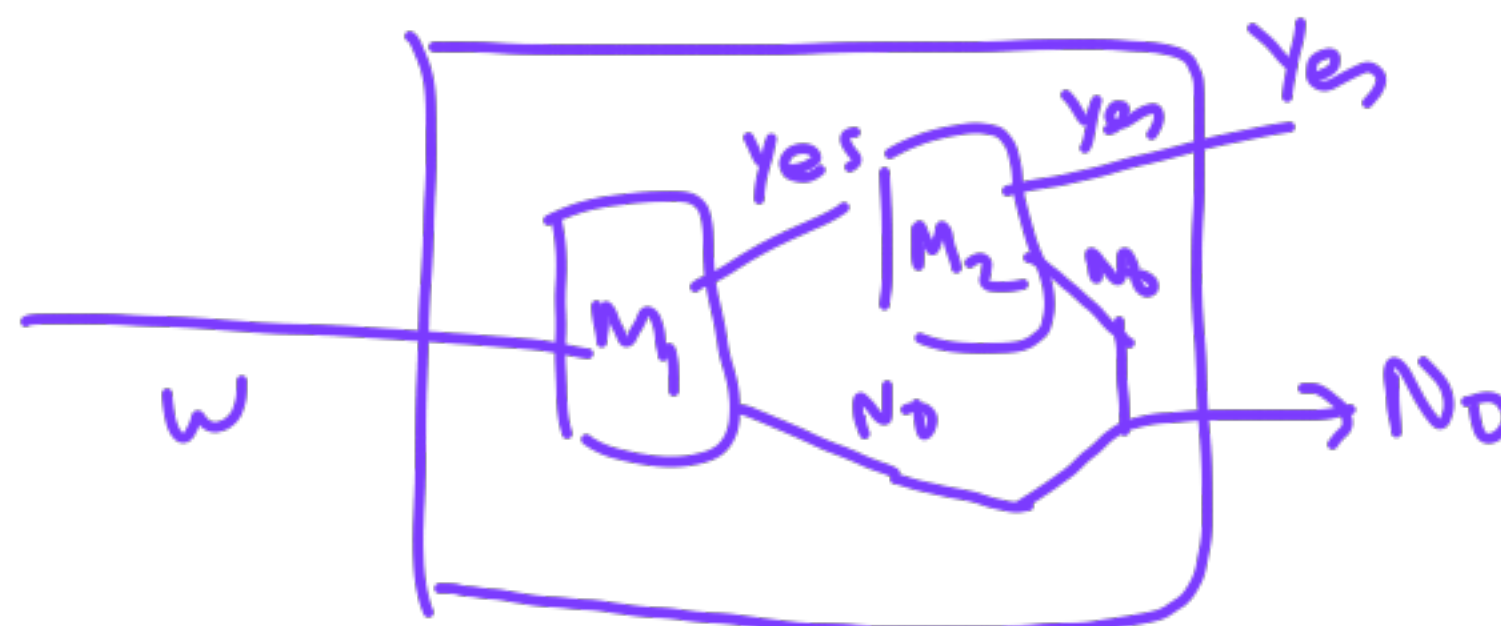
works for r.e.?



Intersection

r.e. $\rightarrow$ h/w

L_1 is rec L_2 is rec
 $L_1 \cap L_2$?



works for r.e.?

H/w $\rightarrow$ If L_1 is r.e. and $\bar{L}_1$ is r.e. then L_1 is recursive

Rec

$L_1 L_2$

L_1^*

L_1^{rev}

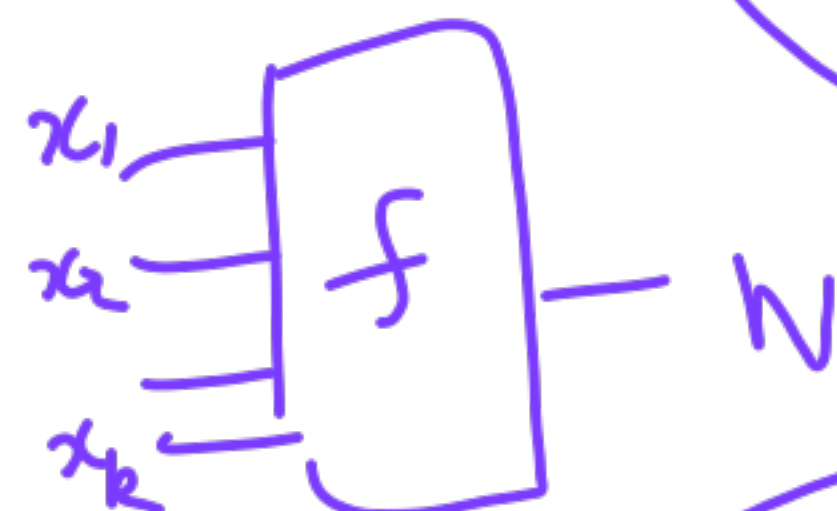
Computations

Can be embedded as Language.

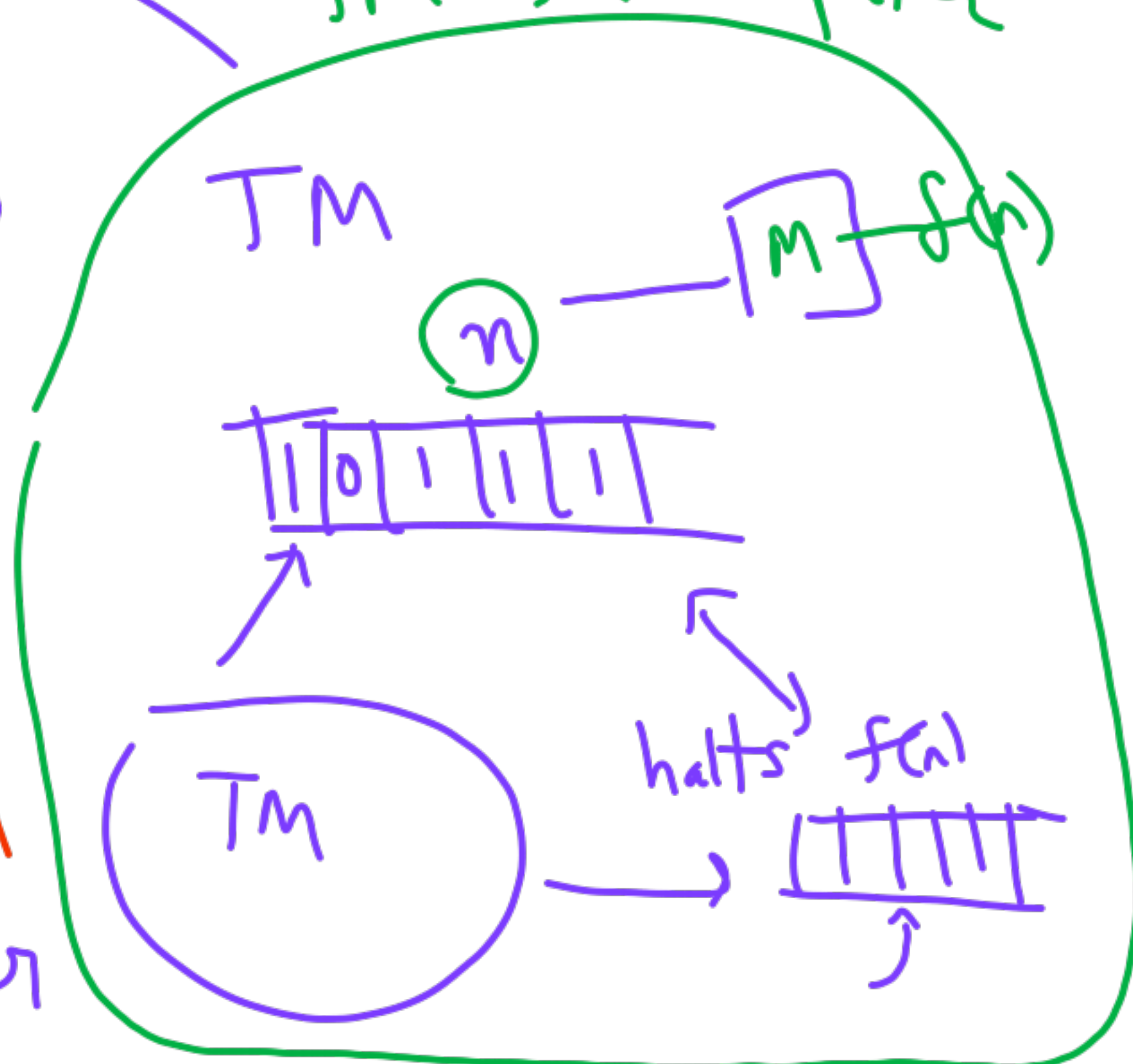
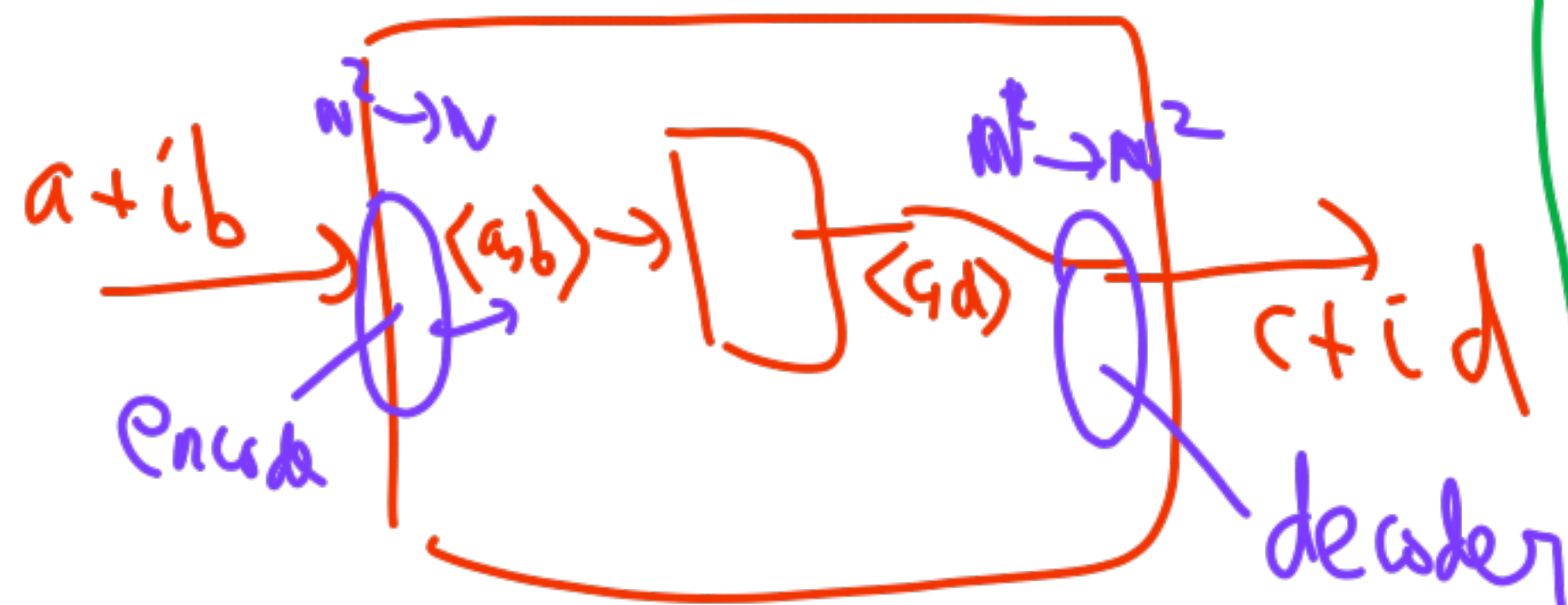
$$a^i b^j c^{i+j}$$

$$a^i b^{2i^2+3i+9}$$

TM as a Computer



$$\langle k, \langle m_1 \dots m_k \rangle \rangle$$



Equivalent computational models.

λ -Expression

λ -calculus

Abstraction

Application

Inductive

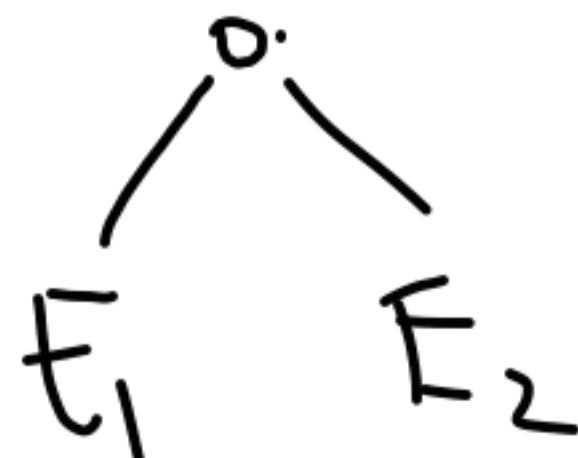
Variable

x, y, z

λx

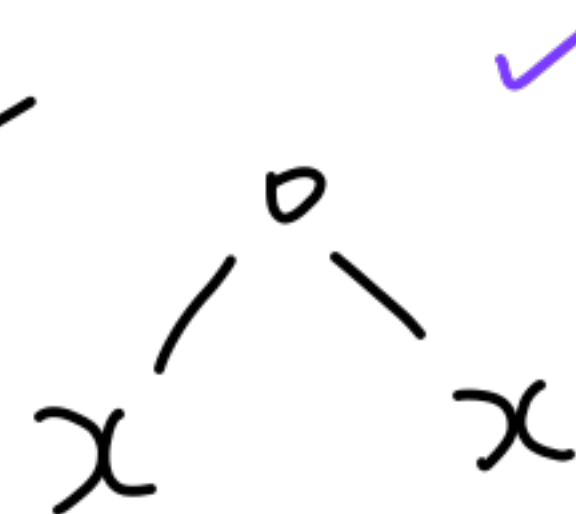
E

any λ -express



β -reductions

Syntax



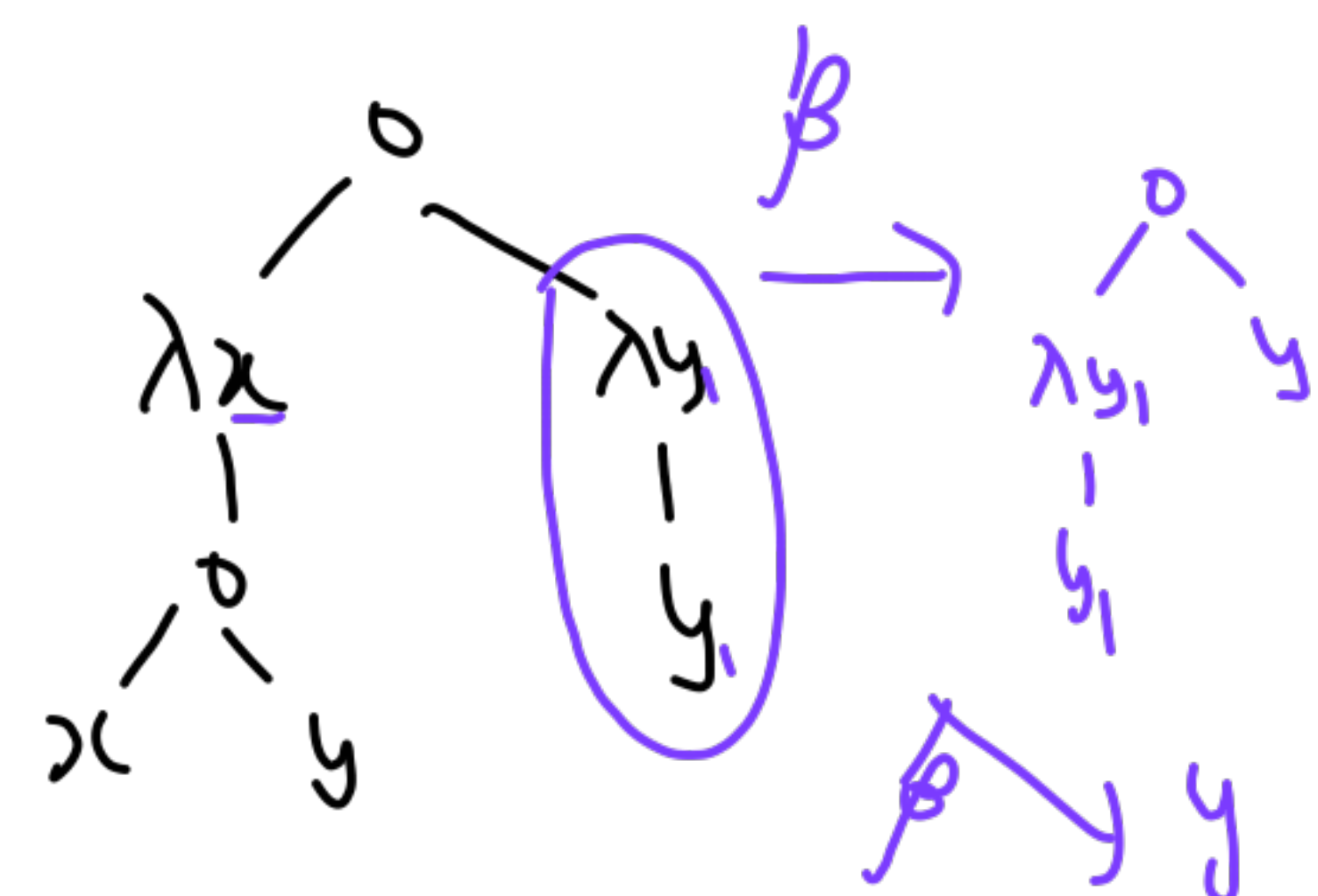
λx

x_1

β

value

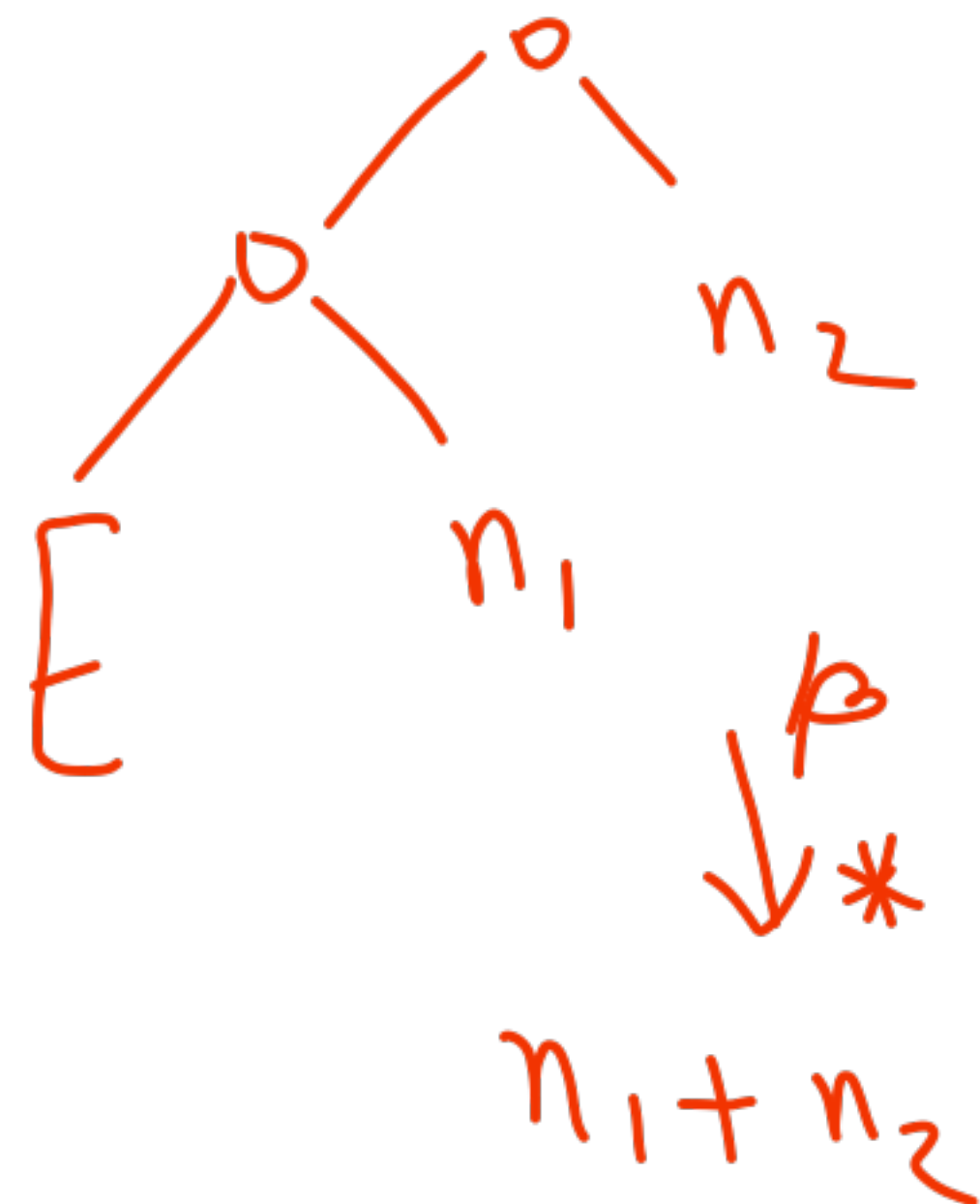
(call by name)



Goal

Numbers

Church
numerals

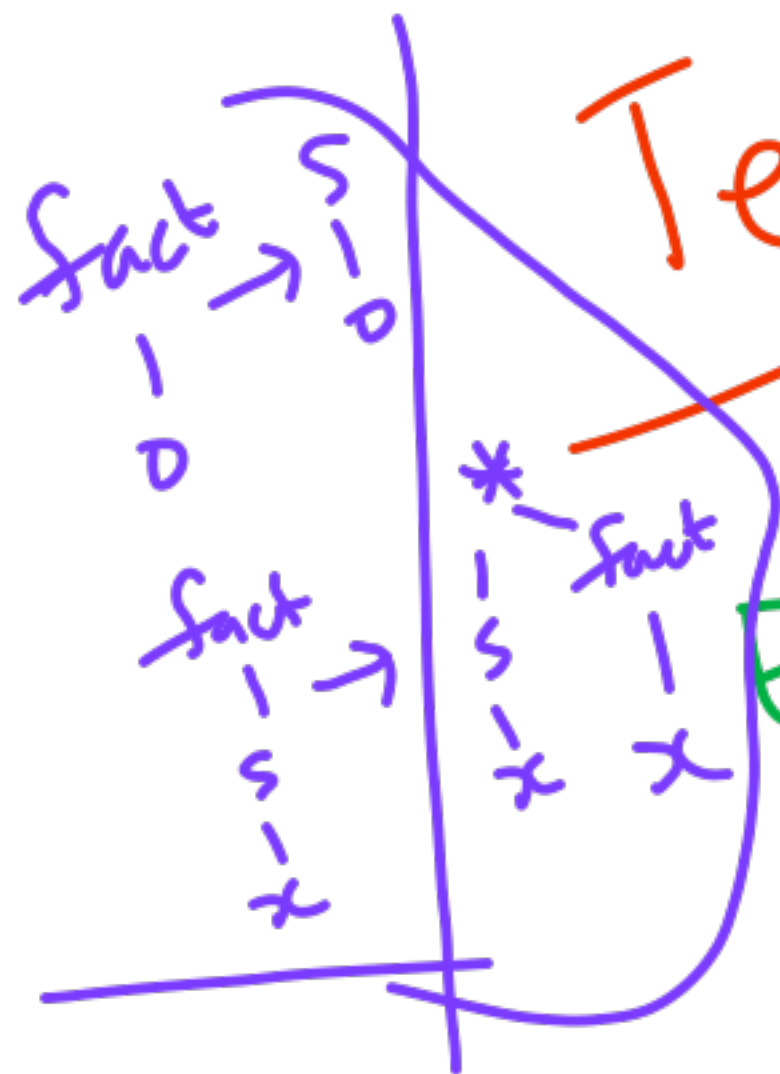


addition
expr

mult
expr



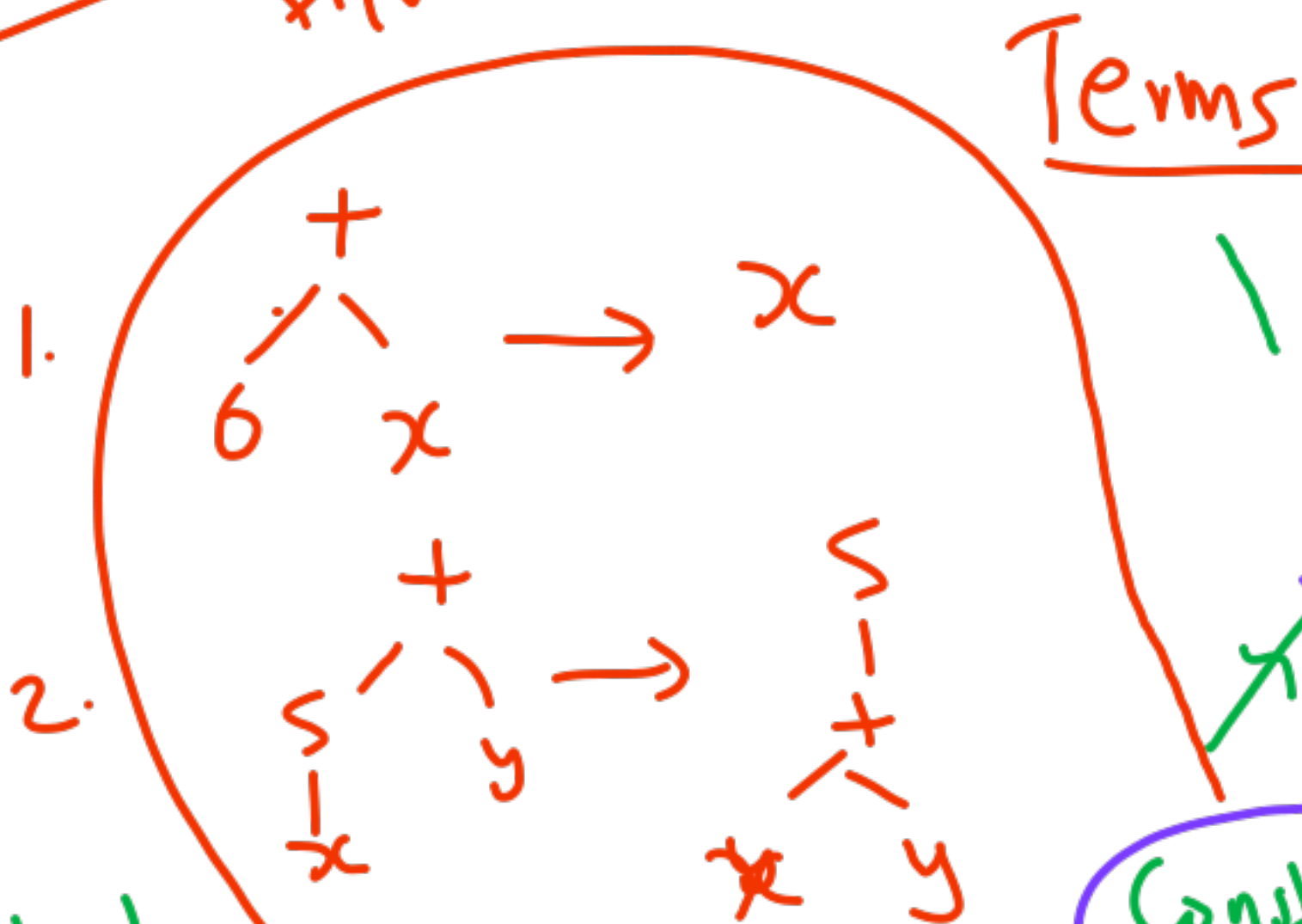
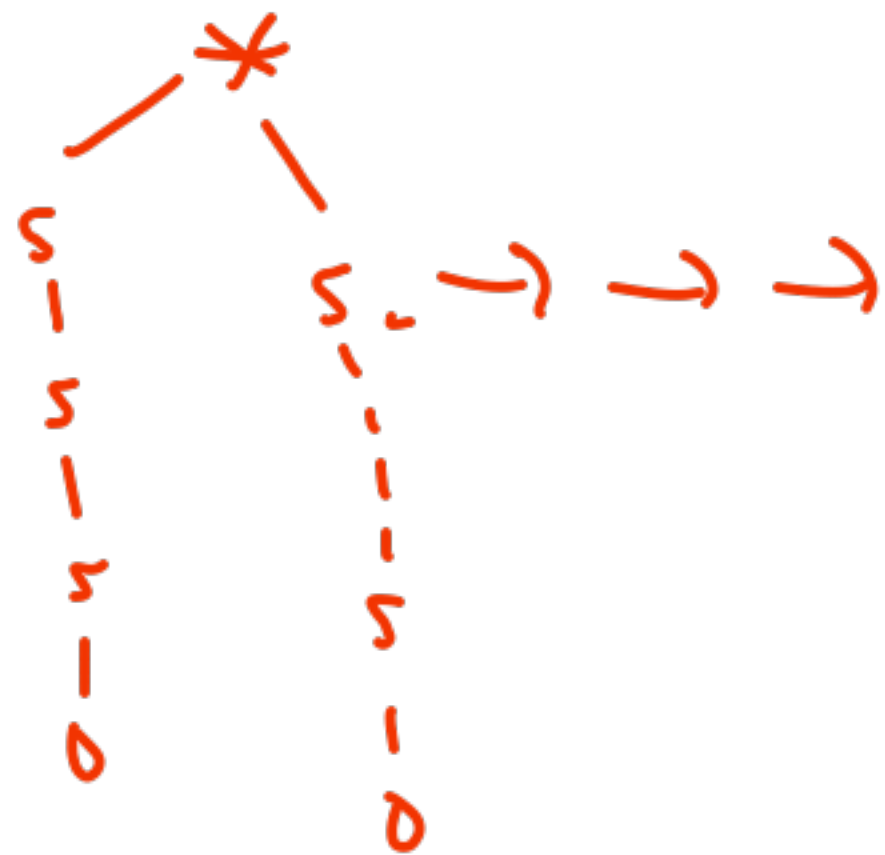
Term Rewriting H/W $\rightarrow$ square



Equality =

"replace equals by equals"

$$3 * 7 = 21$$



Constants

Constructors (domain)

defined by rules

fn. symbol

variable

